C Krithik Kesav

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Education

Birla Institute of Technology and Science Pilani, Hyderabad Campus — Hyderabad, India
Bachelor of Engineering (Hons.) in Mechanical Engineering (2023 – 2027) — *8.8/10.0*
Minor in Data Science — *8.0/10.0*

Technical Skills

Proficiency: Design and Prototyping, Computational Modelling, Data analysis, Machine learning.
Software & Tools: MATLAB, Python, COMSOL, Ansys mechanical, SolidWorks, CREO, SimFlow.

Internships

Summer Research Intern (SURE Program) — IIT Hyderabad
Under the guidance of Dr. Niranjan S Ghaisas — *2026*

- Implemented a BEMT algorithm in MATLAB for propellers with a modification for including the effects of a shroud.
- Validated the algorithm against NASA TN D-4142 ducted rotor as well as against a custom rotor.

Summer Intern — CSIR - NAL
Under the guidance of Dr. Meenu Srivastava — *2025*

- Developed heat transfer models for HVOF thermal spray process to determine optimal cooling rates.
- Developed a model to predict coating layer thickness for different scanning parameters and validated against experimental results.

Projects

Nozzle Shape Optimization — BITS Pilani
Course Project — *Jan 2026 – Apr 2026*

- Built a 2D Method of Characteristics solver to analyze the flow field in a given nozzle contour.
- Implemented Differential Evolution to optimize the nozzle contour to maximize thrust for a given length of a nozzle.

Digital twin for a three-Stage Gearbox — BITS Pilani
Under the guidance of Dr. Sabareeshan Rajasekharan — *Jan 2025 – Ongoing*

- Used Classical ML algorithms to implemet fault diagnosis in a three stage gearbox using accelerometer data.
- 3D printed mounts for sensors and built a live dashboard to display the state of the system using the live RPM and accelerometer sensor feed as well as real time fault classification.

Heat Transfer Analysis of Additive Manufacturing Processes — BITS Pilani
Under the guidance of Dr. Pavan Kumar Penumakala — *Nov 2025 – Ongoing*

- Built multiphysics models in COMSOL to simulate heat transfer, thermal evolution, and material densification under variable process conditions during Selective Laser Sintering (SLS).
- Modelled multi-layer scanning processes in SLS and WAAM to predict the thermal response and thereby optimize process parameters.

Positions of Responsibility

President — 2025 – 2026
Society of Automotive Engineers, BPHC Student Chapter

- Oversaw club administration and managed 2 cross-functional student engineering teams (**60+ members**) for the Baja and Formula projects.
- Raised and managed funding of **INR 6+ lakhs** by pitching through various funding sources, and coordinated the club's administration and campus activities.
- Acted concurrently as **Vehicle Dynamics** Lead for the Baja team, analyzing system-level behavior and managing subsystem integration.

Teaching Assistant — 2025 – 2026
Department of Mechanical Engineering

- TA for the courses ME F212 Fluid Mechanics and ME F314 Design of Machine Elements, which had 180+ students each.
- Conducted doubt-clearing sessions, formulated assignment problems, and helped conduct a project exhibition.